US 20250287066A1

(19) **United States**

(12) **Patent Application Publication**

Huddar et al.

(10) **Pub. No.: US 2025/0287066 A1**

(43) **Pub. Date: Sep. 11, 2025**

(54) **4K CONTENT MEMORY FRAGMENTATION VIEWING COUNTERMEASURES**

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(21) Appl. No.: **18/601,798**

(22) Filed: **Mar. 11, 2024**

Publication Classification

(51) **Int. Cl.**

H04N 21/44 (2011.01)

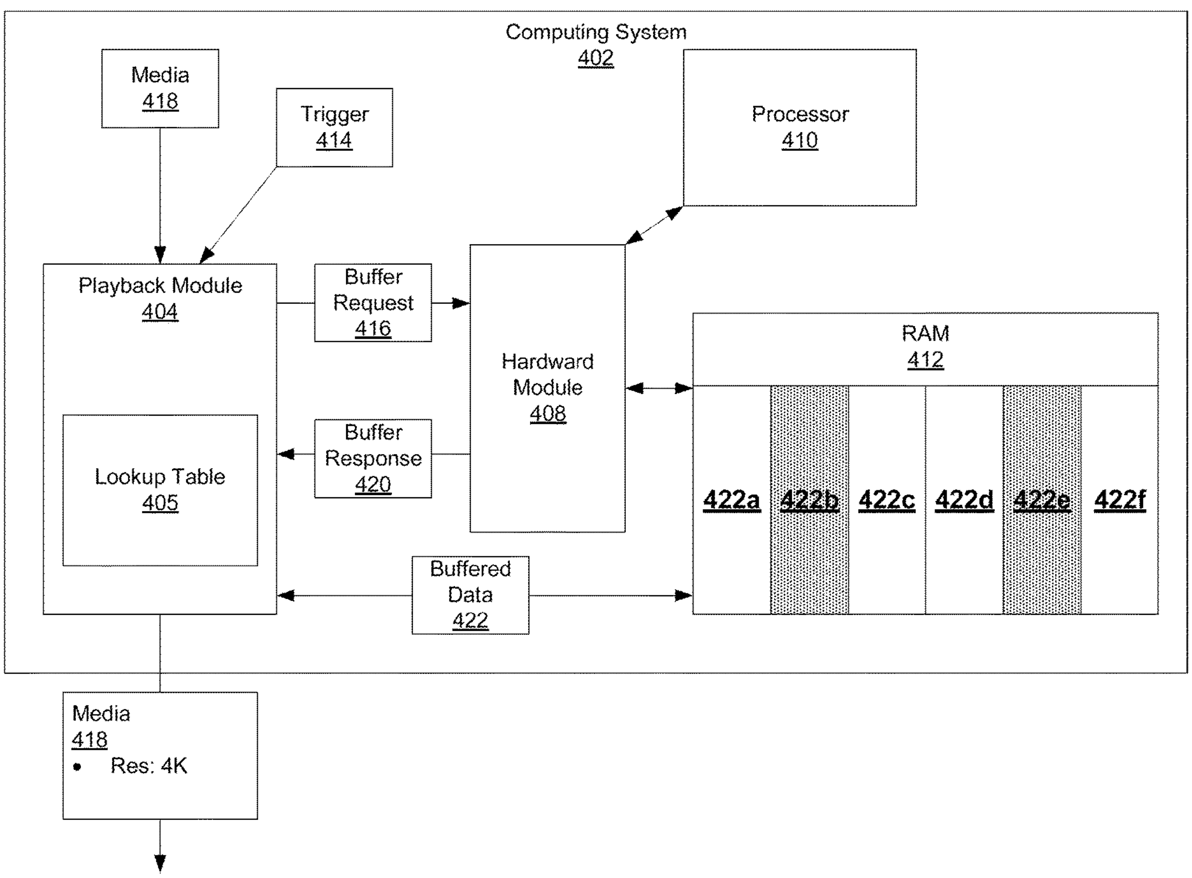
H04N 21/442 (2011.01)

(52) **U.S. Cl.**

CPC . ***H04N 21/44004*** (2013.01); ***H04N 21/44209*** (2013.01)

(57) **ABSTRACT**

A method may include receiving, by a playback module of a computing system, a trigger to increase a resolution of media to be displayed from a first resolution to a second resolution. The method may include determining, by the playback module of the computing system, a minimum buffer associated with the second resolution. The method may include transmitting, by the playback module of the computing system, a buffer query to a hardware module of the computing system, the buffer query indicating the minimum buffer. The method may include receiving, by the playback module of the computing system, a buffer response from the hardware module of the computing system, the buffer response indicating whether the minimum buffer is available. The method may include causing, by the playback module of the computing system, the media to be displayed at the second resolution.



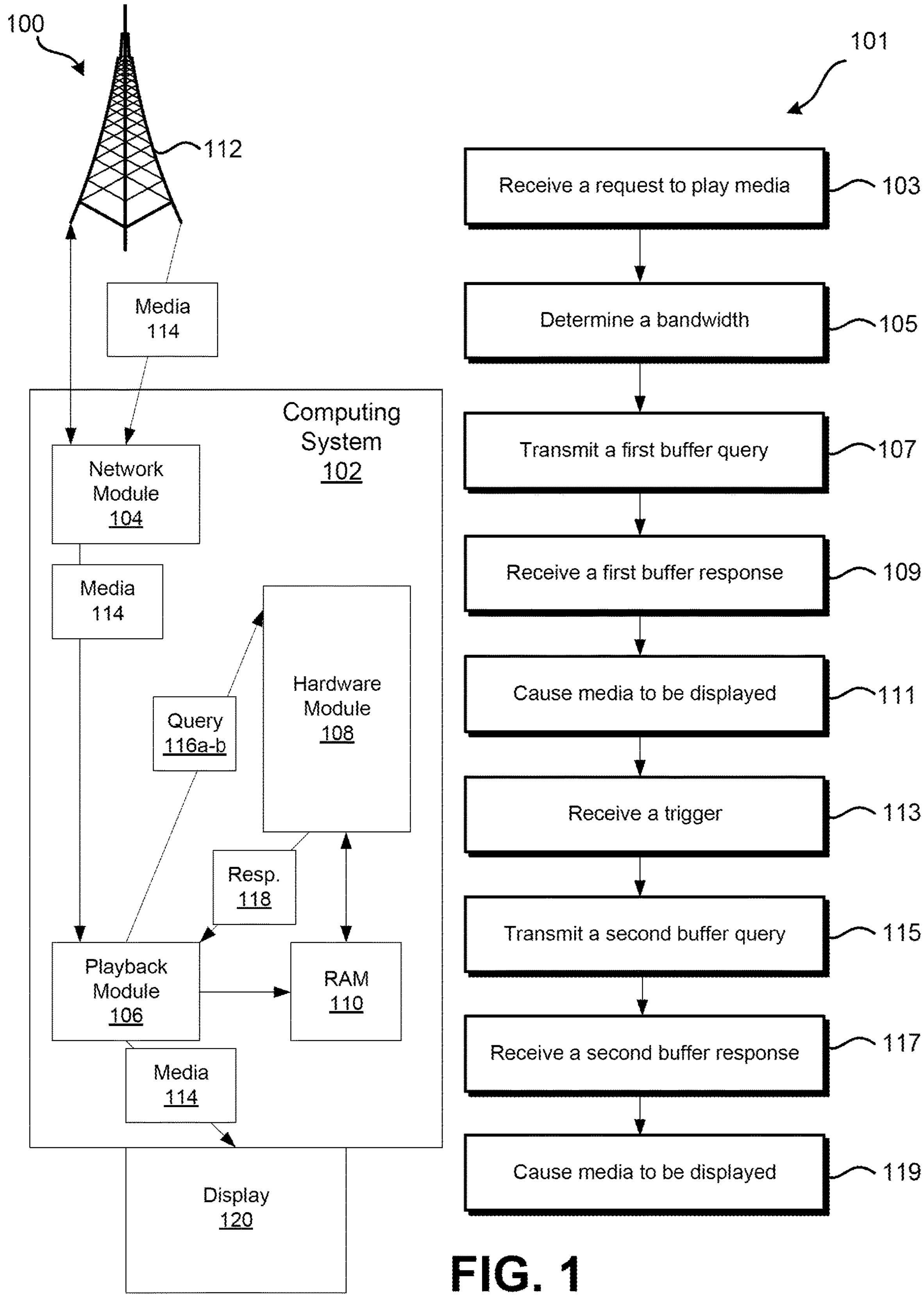

100
112
Media
114
Computing
System
102
Network
Module
104
Media
114
Hardware
Module
108
Query
116a-b
Resp.
118
Playback
Module
106
RAM
110
Media
114
Display
120
101
Receive a request to play media
103
Determine a bandwidth
105
Transmit a first buffer query
107
Receive a first buffer response
109
Cause media to be displayed
111
Receive a trigger
113
Transmit a second buffer query
115
Receive a second buffer response
117
Cause media to be displayed
119

FIG. 1

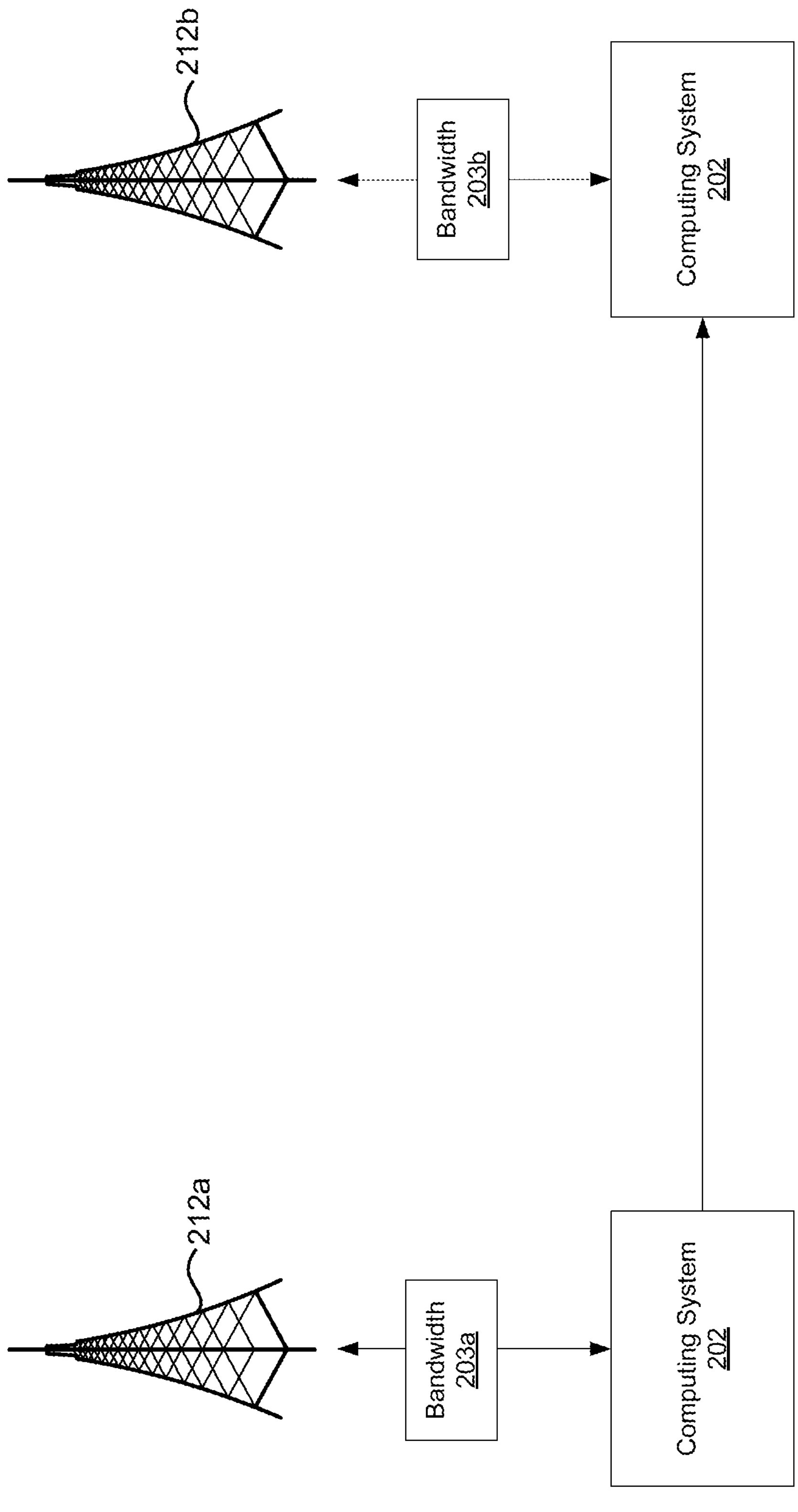

212b
Bandwidth
203b
Computing System
202
212a
Bandwidth
203a
Computing System
202

FIG. 2

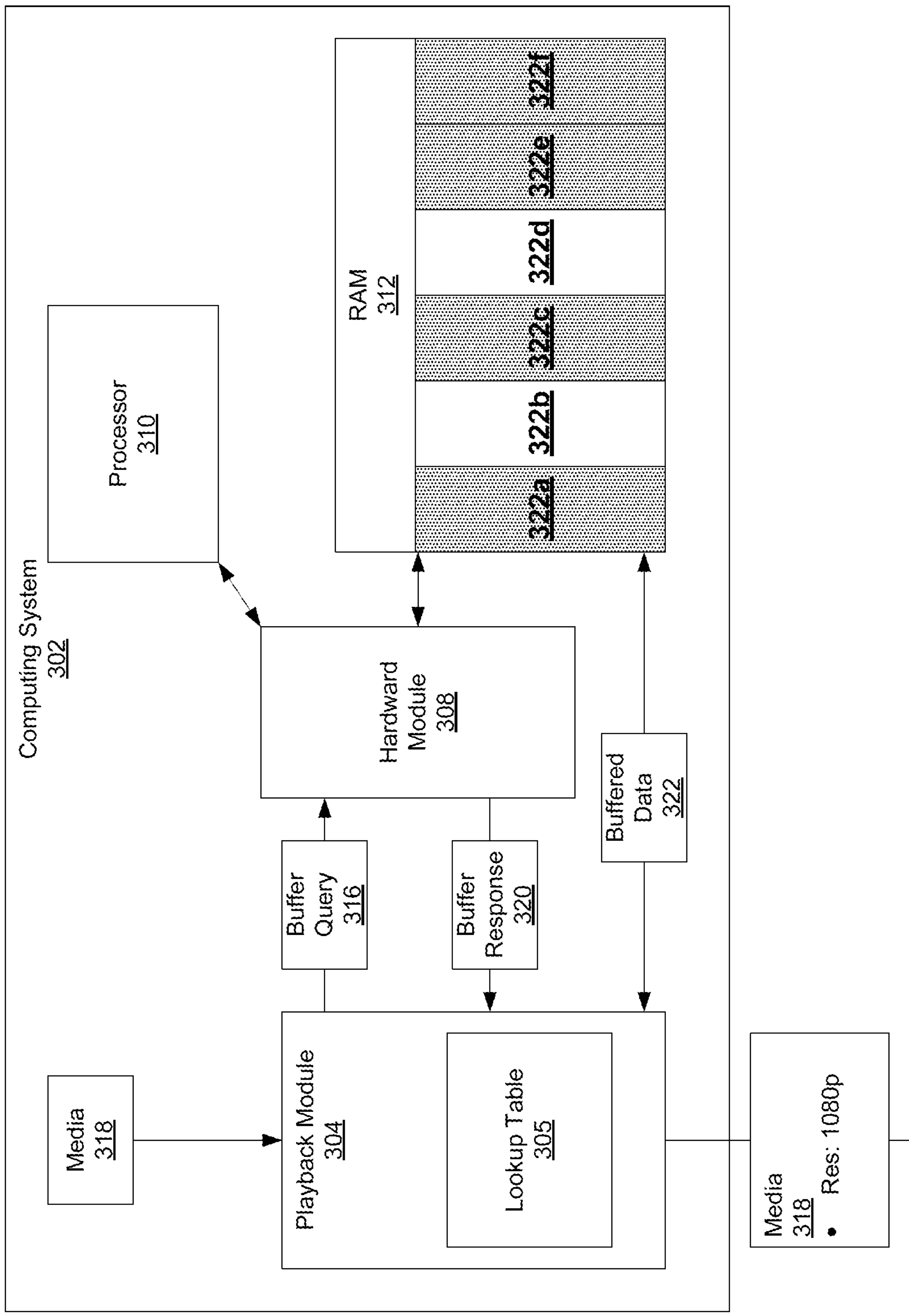

Computing System 302
Processor 310
RAM 312
322a
322b
322c
322d
322e
322f
Hardward Module 308
Buffer Query 316
Buffer Response 320
Buffered Data 322
Media 318
Playback Module 304
Lookup Table 305
Media 318
Res: 1080p

FIG. 3

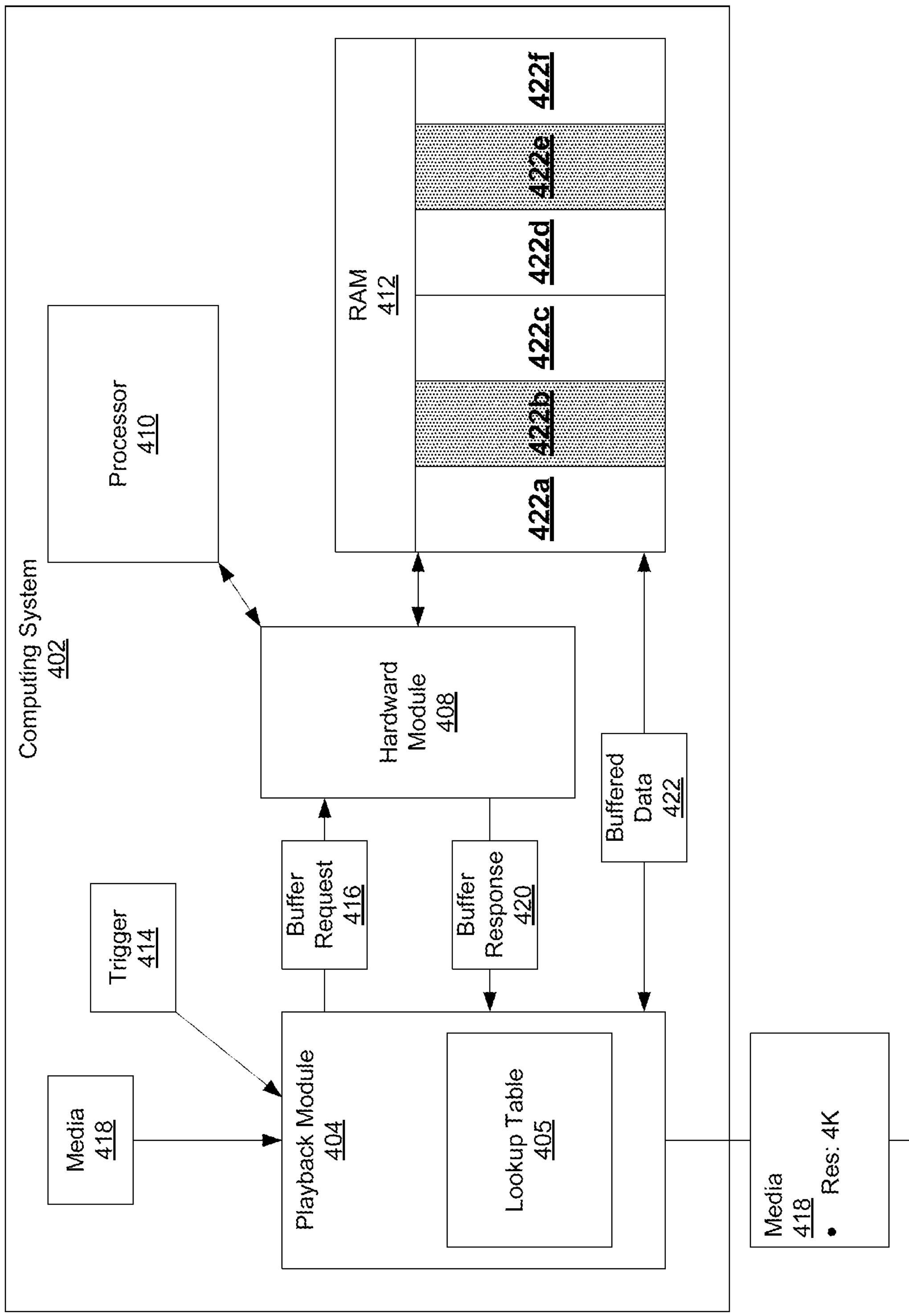

Computing System
402
Processor
410
RAM
412
422a
422b
422c
422d
422e
422f
Hardward Module
408
Buffered Data
422
Trigger
414
Buffer Request
416
Buffer Response
420
Media
418
Playback Module
404
Lookup Table
405
Media
418
• Res: 4K

FIG. 4

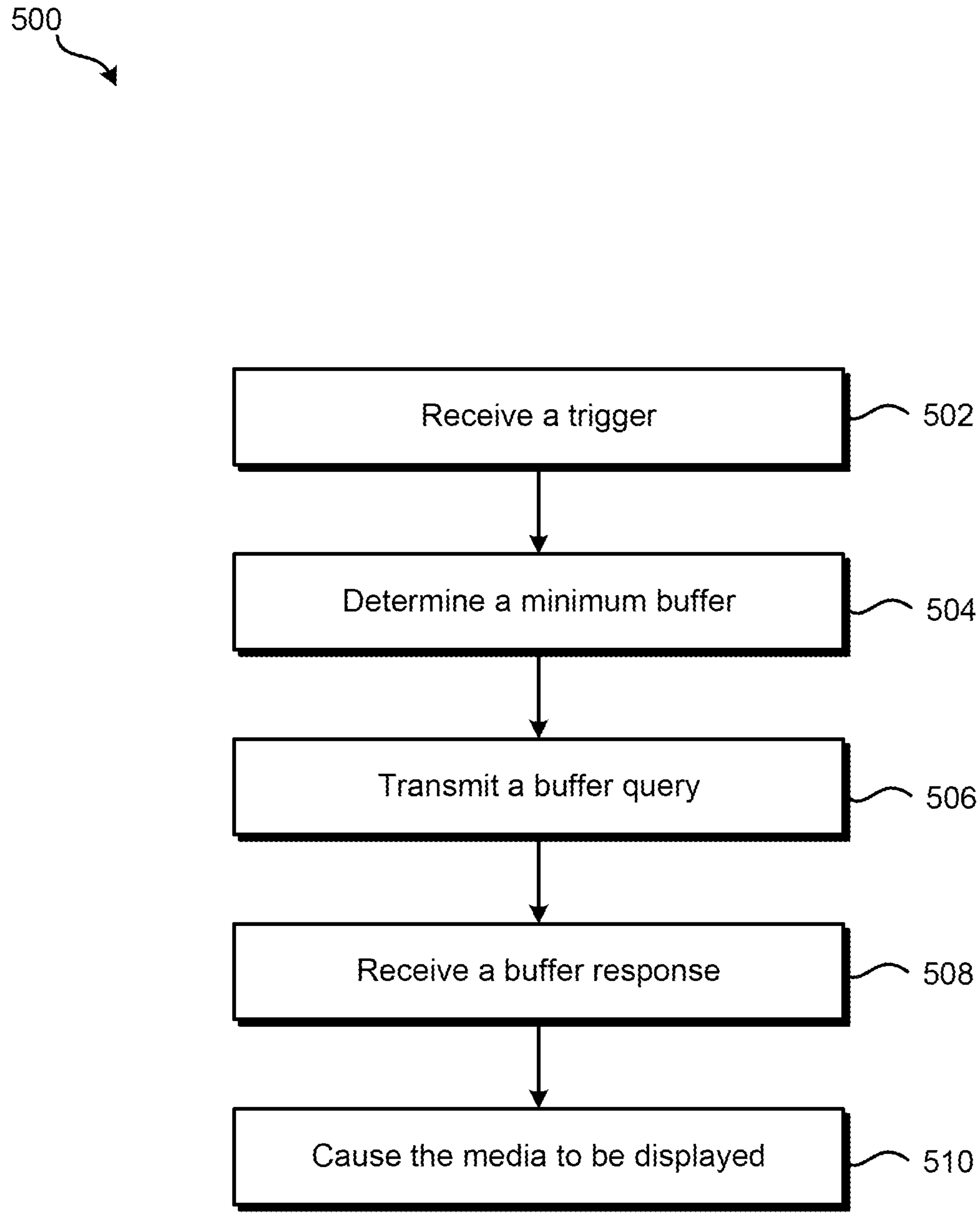

500
Receive a trigger
502
Determine a minimum buffer
504
Transmit a buffer query
506
Receive a buffer response
508
Cause the media to be displayed
510

FIG. 5

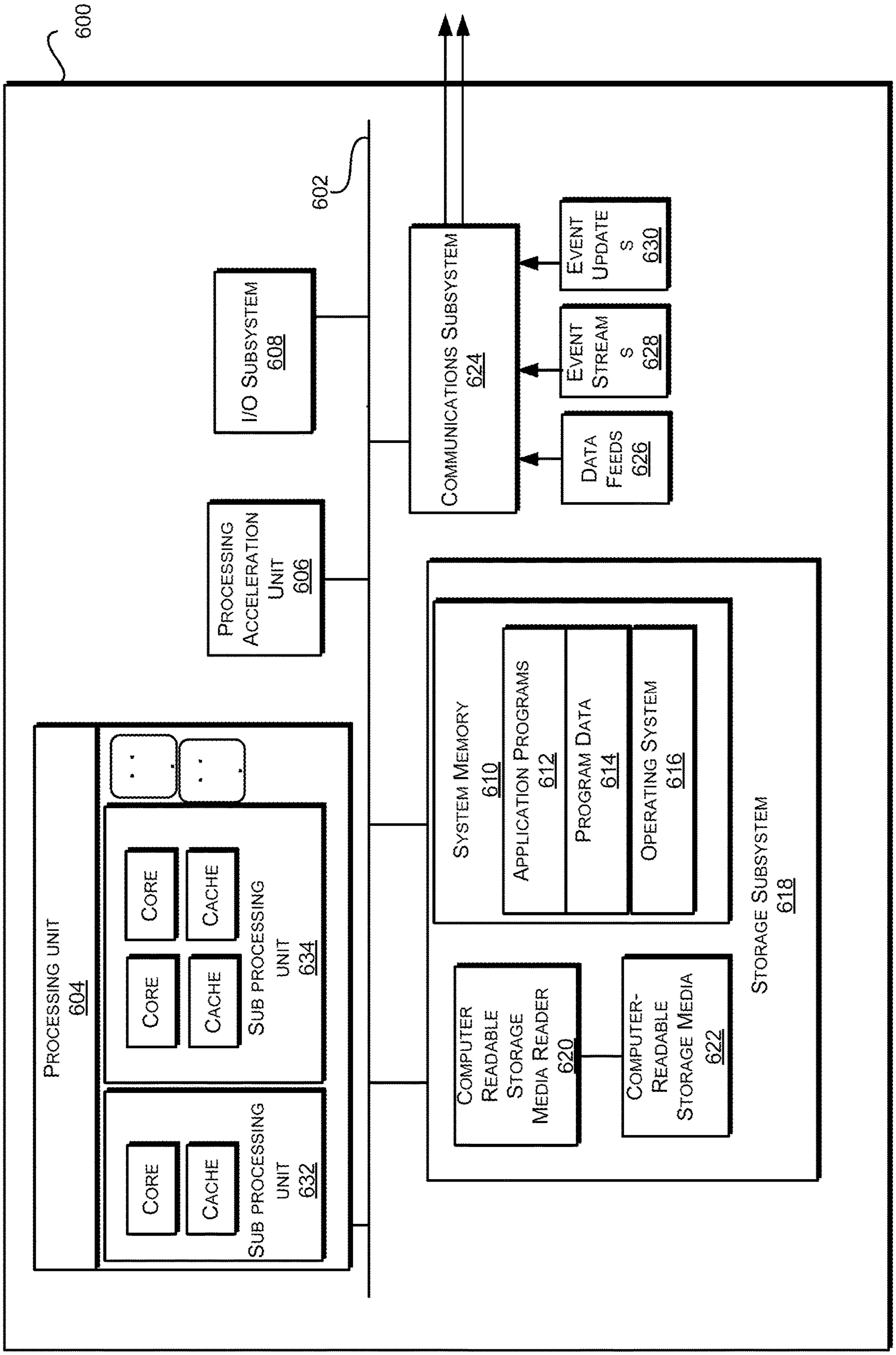

600
602
Processing unit
604
Core
Cache
Sub processing unit
632
Core
Cache
Core
Cache
Sub processing unit
634
Processing Acceleration Unit
606
I/O Subsystem
608
Communications Subsystem
624
Data Feeds
626
Event Stream s
628
Event Update s
630
System Memory
610
Application Programs
612
Program Data
614
Operating System
616
Computer Readable Storage Media Reader
620
Computer-Readable Storage Media
622
Storage Subsystem
618

FIG. 6

4K CONTENT MEMORY FRAGMENTATION VIEWING COUNTERMEASURES

BACKGROUND

[0001] Streaming media via a network to a computing system or device requires a certain amount of bandwidth from the network in order to transmit the media to the computing system. The computing system must also have the resources to play the media at any given resolution. As resolutions have increased, the resources needed to satisfactorily play the media on the computing system. The increased resolutions have caused some computing systems to fragment the media, leading to errors in playback.

BRIEF SUMMARY

[0002] A method may include receiving, by a playback module of a computing system, a trigger to increase a resolution of media to be displayed from a first resolution to a second resolution. The method may include determining, by the playback module of the computing system, a minimum buffer associated with the second resolution. The method may include transmitting, by the playback module of the computing system, a buffer query to a hardware module of the computing system, the buffer query indicating the minimum buffer. The method may include receiving, by the playback module of the computing system, a buffer response from the hardware module of the computing system, the buffer response indicating whether the minimum buffer is available. In response to the buffer response indicating that the minimum buffer is available, the method may include causing, by the playback module of the computing system, the media to be displayed at the second resolution.

[0003] In some embodiments, in response to the buffer response indicating that the minimum buffer is unavailable, the method may include causing, by the playback module of the computing system, the media to be displayed at the first resolution. The first resolution may be at least one of 1080p or 720p, and the second resolution may be 4 k. The minimum buffer corresponds to at least one continuous block of random access memory. The buffer query is transmitted to the hardware module of the computing system using an application programming interface.

[0004] In some embodiments, the method may include determining, by the computing system, a first bandwidth for wireless communications via a wireless network. The method may include detecting, by the computing system, an increase in the first bandwidth for wireless communications to a second bandwidth. The method may include determining, by the playback module of the computing system, that the second bandwidth is above a certain threshold such that the media may be displayed at the second resolution. The computing system may transmit the buffer request in response to determining that the second bandwidth is above the certain threshold. The trigger may include a change in bandwidth available to the computing system via a network. In some embodiments, the method may include in response to the buffer response indicating that the minimum buffer is available, receiving, from the hardware module of the computing system, a location of a continuous block of memory within a computer-readable memory corresponding to the minimum buffer. The method may include causing, by the playback module, data associated with the media to be stored in at least a portion of the continuous block of memory.

[0005] A system may include one or more processors, a hardware module configured to access data indicating a status of one or more hardware components of the system, a playback module, configured to cause media to be displayed by the system, and a non-transitory computer-readable medium may including instructions. When executed by the one or more processors, the instructions may cause the system to perform operations. According to the operations, the system may receive, by the playback module, a trigger to increase a resolution of media to be displayed from a first resolution to a second resolution. The system may determine, by the playback module, a minimum buffer associated with the second resolution. The system may transmit, by the playback module, a buffer query to the hardware module, the buffer query indicating the minimum buffer; receive, by the playback module, a buffer response from the hardware module of the computing system, the buffer response indicating whether the minimum buffer is available. In response to the buffer response indicating that the minimum buffer is available, the system may cause, by the playback module, the media to be displayed at the second resolution.

[0006] In some embodiments, the first resolution may be at least one of 1080p or 720p, and the second resolution may be 4 k. The buffer query may include respective buffer sizes associated with at least one of the first resolution or the second resolution. The first resolution may be determined at least in part by an available bandwidth of a network. The system may further performs operations to determine a first bandwidth for wireless communications via a wireless network. The system may detect an increase in the first bandwidth for wireless communications to a second bandwidth. The system may determine, by the playback module, that the second bandwidth is above a certain threshold such that the media may be displayed at the second resolution. The playback module may transmit the buffer query in response to determining that the second bandwidth is above the certain threshold.

[0007] A non-transitory computer-readable medium may include instructions that, when executed by one or more processors, cause the one or more processors to perform operations. The operations may include receiving, by a playback module of a computing system, a trigger to increase a resolution of media to be displayed from a first resolution to a second resolution. The operations may include determining, by the playback module of the computing system, a minimum buffer associated with the second resolution. The operations may include transmitting, by the playback module of the computing system, a buffer query to a hardware module of the computing system, the buffer query indicating the minimum buffer. The operations may include receiving, by the playback module of the computing system, a buffer response from the hardware module of the computing system, the buffer response indicating whether the minimum buffer is available. The operations may include, in response to the buffer response indicating that the minimum buffer is available, causing, by the playback module of the computing system, the media to be displayed at the second resolution.

[0008] In some embodiments, the minimum buffer corresponds to a plurality of continuous blocks of random access memory. The buffer query may be transmitted to the hardware module of the computing system using an application programming interface. The operations may include determining, by the computing system, a first bandwidth for wireless communications via a wireless network. The operations may include detecting, by the computing system, an increase in the first bandwidth for wireless communications to a second bandwidth. The operations may include determining, by the playback module of the computing system, that the second bandwidth is above a certain threshold such that the media may be displayed at the second resolution.

[0009] In some embodiments, the operations may include, in response to the buffer response indicating that the minimum buffer is available, receiving, from the hardware module of the computing system, a location of a continuous block of memory within a computer-readable memory corresponding to the minimum buffer. The operations may include causing, by the playback module, data associated with the media to be stored in at least a portion of the continuous block of memory.

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] FIG. **1** illustrates a system and a process for reducing memory fragmentation during media playback, according to certain embodiments.

[0011] FIG. **2** illustrates a system including a computing system and networks, according to certain embodiments.

[0012] FIG. **3** illustrates a system for reducing memory fragmentation during media playback, according to certain embodiments.

[0013] FIG. **4** illustrates system for reducing memory fragmentation during media playback, according to certain embodiments.

[0014] FIG. **5** illustrates a flowchart of a method for reducing memory fragmentation during media playback, according to certain embodiments.

[0015] FIG. **6** illustrates an exemplary computer system **600**, according to certain embodiments.

DETAILED DESCRIPTION

[0016] Streaming video has become a popular way to consume media. As the resolution of displays has increased, the file size of video has also increased. The increased file size associated with high resolution media (e.g., video) may require higher bandwidth to deliver the media to a computing system or device. Some wireless networks may provide enough bandwidth to deliver the appropriate files needed to play the media at a high resolution (e.g., 4K, 1080p, etc.). Furthermore, the computing system itself may also utilize more hardware resources to properly play back the media. For example, in order to protect against frame drop, video fragmentation, and other undesirable effects, the computing system may have an in-memory buffer. The in-memory buffer may be allocated to store some or all of the media in order to render the media at the desired resolution. However, even within the same wireless network, the network bandwidth available to a particular device may rise and fall depending on network conditions, location, user volume, and other such factors. Thus, when a user is playing the media at one resolution, a change in bandwidth may make a higher resolution possible. The computing system, however, may not be able to provide the resources needed in order to play the media at the higher resolution.

[0017] For example, a computing system connected to a network may request media to be streamed and displayed by the computing system and/or by a different display. The request may include a request to play the media at the highest possible resolution. The computing system may request the media to be downloaded according to the highest possible resolution, but the bandwidth of the network may be too low to permit streaming the media at the requested resolution. The computing system may then select the highest available resolution, and allocate resources such as an in-memory buffer and/or processor resources according to the highest available resolution. After receiving the media via the network, the computing system may play the media at the highest possible resolution.

[0018] At some point later, the computing system may detect that the network (or a different network) has enough bandwidth to play at the highest possible resolution. However, because the computing system allocated resources for the highest available resolution, the computing system may not be able to display the media at the highest possible resolution. For example, the highest available resolution may only require a continuous buffer of 6 MB. The highest possible resolution may require a continuous buffer of 21 MB. If the computing system attempted to play the media at the highest possible resolution, the media may appear to freeze, drop video frames, audio, and have other undesired effects. Thus, even if a change in the bandwidth enables a switch from one resolution to a higher resolution, there may be significant problems when trying to display the media at the higher resolution.

[0019] One solution to this problem may be to query the computing system in order to detect available continuous blocks of memory before displaying media at a higher resolution. A computing system (sometimes "computing device") may receive a request (e.g., a user input) to stream media such as a video. The computing system may be configured to process and render the media in any number of resolutions (e.g., 720p, 1080p, 4K, 8K, etc.). The computing system may determine that a wireless connection to a wireless network has a bandwidth capable of supporting streaming a first resolution (e.g., 1080p). A playback module of the computing system may then request a graphics buffer associated with the first resolution (e.g., 3 continuous blocks of RAM, where each block is 6 MB) from a hardware module of the computing system. The hardware module may then determine a total amount of memory available and a total amount of continuous blocks available. If the hardware module determines that the requested memory is available-both the total amount of memory and the continuous blocks—the hardware module may cause the graphics buffer to be allocated within the RAM. The hardware module may then provide a response to the playback module indicating the location of the memory to be used as the graphics buffer. The playback module may then receive data associated with the media from the network, cause at least a portion of the data to be stored in the graphics buffer, and cause the media to be displayed (either on an on-board display or some other display).

[0020] Later, the computing system may detect that the bandwidth provided by the wireless connection has increased such that the media may be streamed at a second resolution (e.g., 4K). Before requesting the media via the network, the playback module may transmit a second request for the graphics buffer to the hardware module, indicating an amount of total memory and continuous blocks thereof associated with the second resolution (e.g., 84 MB total memory, in continuous blocks of 21 MB). The hardware module may then determine the total amount of available memory and the amount of the available memory that is in continuous blocks. If the hardware module determines that the computing system has the memory available to fulfil the second request for the graphics buffer, the hardware module may allocate the memory within the RAM for the graphics buffer. The hardware module may then transmit a response to the playback module indicating the updated location(s) of the graphics buffer. The playback module may then request the media at the second resolution via the network. Upon receiving data associated with the media, the playback module may cause at least a portion of the data to be stored in the graphics buffer and cause the media to be displayed (either on an on-board display or some other display). If, by contrast, the hardware module determines that the requested graphics buffer is unavailable, the hardware module may transmit a response to the playback module indicating that the requested graphics buffer is unavailable. The playback module may then continue causing the media to be displayed at the first resolution.

[0021] FIG. **1** illustrates a system **100** and a process **101** for reducing memory fragmentation during media playback, according to certain embodiments. The system **100** may include a computing system **102**. The computing system **102** may include a network module **104**, a playback module **106**, a hardware module **108**, RAM **110**, and a display **120**. The computing system **102** may be a unitary device, such as a mobile phone tablet, laptop, etc. Alternatively, some of the components of the computing system **102** may be separate from other components. For example, the display **120** may be a monitor, display, or other computing device, connected to the computing system **102** via a wired or wireless connection (e.g., WiFi, Bluetooth, etc.).

[0022] The network module **104** may include one or more hardware and/or software components. The network module **104** may be configured to communicate with one or more networks such as cellular network (e.g., 3G, 4G, 5G, etc.), WiFi, a wired connection, or any other suitable network. The network module **104** may transmit and receive communications from a network (e.g., a network **112**) and/or other components of the computing system **102**. The network module **104** may also be configured to determine one or more properties associated with the network, such as available bandwidth, signal strength, and other properties.

[0023] The playback module **106** may include hardware and/or software components and be configured to render and display media via components of the computing system **102**. The playback module **106** may enable the media to be displayed at one or more resolutions, such as 720p, 1080p, 4K, 8K, and any other resolution. The playback module **106** may determine the one or more possible resolutions based on the particular components of the computing system **102** and/or the components thereof. For example, the display **120** may only be capable of displaying the media at resolutions of 1080p or lower. In that case, the playback module **106** may only be configured to process media at resolutions of 1080p or lower. In other examples, the playback module **106** may be configured to process media at any or all resolutions regardless of the capabilities of the computing system **102** and/or the components thereof.

[0024] The hardware module **108** may be configured to determine one or more properties of the computing system **102**. The hardware module **108** may include one or more software and/or hardware components. In some embodiments, the hardware module **108** may be included in an operating system of the computing system **102**. The hardware module **108** may monitor various components of the computing system **102** continuously during operation and/or may determine properties of the various components in response to a request. The hardware module **108** may also be configured to cause the properties of the various components. For example, the hardware module **108** may cause memory (e.g., the RAM **110**) to be allocated to tasks being performed by the computing system **102**. The hardware module **108** may also be configured to cause processor time to be allocated to the tasks being performed by the computing system.

[0025] The RAM **110** may be a single memory chip within the computing system **102** or may be multiple memory chips. In some embodiments, the RAM **110** may be dedicated to a specific purpose, such as graphics processing. In other embodiments, portions the RAM **110** may be allocated (e.g., by the hardware module **108**) to the specific purpose (s). The RAM **110** may be static RAM (SRAM), dynamic RAM (DRAM), LPDDR4, LPDDR4x, and any other suitable memory type. The RAM **110** may have a total amount of memory (e.g., 8 GB, 16 GB, 32 GB, etc.) divided into logical and/or physical blocks. Each of the blocks may include a plurality of bits, identifiable by addresses indicating the location of one or more of the plurality of bits. Each block may therefore be identified by a series of corresponding addresses. The blocks themselves may be of any size. For example, a first block may include 4 MB of unused bits and be indicated by a series of addresses corresponding the unused bits. A second block may include 4 MB of bits in use (e.g., unavailable bits), and be indicated by a series of addresses corresponding to the unavailable bits. The display **120** may include a screen integrated into the computing system **102** (e.g., the screen of a mobile device) or may be a display connected to the computing system **102** (e.g., a monitor connected to a set top box or other such device). The display **120** may be configured to display media at a maximum resolution (e.g., 8K).

[0026] At step **103**, the computing system **102** may receive a request to play media **114**. The request may be a user request to stream the media **114** via the network **112**. The request may be received via the playback module **106** and/or may be transmitted to the playback module **106** via another component of the computing system **102**, such as the operating system. The request may include a desired resolution. The desired resolution may the maximum resolution of the display **120**, or may be a preset resolution based on a predetermined setting.

[0027] At step **105**, the network module **104** may determine a bandwidth available to the computing system **102** via the network **112**. The network module **104** may determine the bandwidth based on the request to play the media **114**. For example, the request to play the media **114** may trigger the network module **104** to determine the available bandwidth. The network module **104** may then indicate the available bandwidth to the playback module **106**. The playback module **106** may then determine a first resolution based on the available bandwidth. In other embodiments, the playback module **106** may query the network module **104**

for the available bandwidth, indicating a bandwidth associated with the desired resolution. The network module **104** may then respond with the available bandwidth, and/or indicate that the desired resolution may be supported by the available bandwidth. In other examples, the network module **104** may respond with an available bandwidth less than that associated with the desired bandwidth. The playback module **106** may then determine the first resolution based on the available bandwidth.

[0028] At step **107**, the playback module **106** may transmit a first buffer query **116***a* to the hardware module. The playback module **106** may transmit the first buffer query **116***a* to the hardware module **108** via an application programming interface (API). The first buffer query **116***a* may indicate a total amount of memory and/or continuous blocks of memory associated with the first resolution. For example, the first resolution may be 1080p. The playback module **106** may determine an amount of the media **114** (e.g., 2 seconds of playback, 3 seconds, etc.) to be buffered in order to display the media **114** properly. The playback module **106** may then determine that the amount of media **114** to be buffered at 1080p is 24 MB of memory in 6 MB continuous blocks (e.g., 8 seconds of buffered media **114**). The first buffer query **116***a* may then indicate a request for 24 MB of the RAM **110** in 6 MB continuous blocks.

[0029] In other embodiments, the first buffer query **116***a* may indicate multiple buffer sizes. For example, the playback module **106** may determine that the available bandwidth may only support 1080p, but that other components of the computing system **102** are generally able to support a higher resolution (e.g., 4K). The first buffer query **116***a* may then indicate the various buffer sizes. For example, the playback module **106** may determine that the amount of the media **114** to be buffered at the higher resolution is 84 MB of memory in 21 MB continuous blocks (e.g., 8 seconds of the media **114**). The buffer query **116** may then indicate a request for 84 MB in 21 MB continuous blocks. The buffer query **116** may also indicate a secondary request for the buffer size associated with the first resolution, and/or other buffer sizes associated with other resolutions. The hardware module **108** may then iteratively process the requests included in the first buffer query **116***a* until a buffer size is found that is available within the RAM **110**. In any case, the hardware module **108** may then cause the RAM **110** to allocate continuous blocks of memory to fulfil the first buffer query **116***a*. In some embodiments, the hardware module **108** may also query one or more processors of the computing system **102** to determine a capacity needed to allocate the amount of RAM **110** needed for the buffer.

[0030] At step **109**, the playback module **106** may receive a buffer response **118** from the hardware module **108**. The buffer response **118** may indicate that the indicated amount of memory for the first resolution has been allocated within the RAM **110**. The buffer response **118** may also include the addresses indicating the location of the allocated memory (now, the "graphics buffer"). In embodiments where the first buffer query **116***a* indicates multiple buffer sizes, the buffer response **118** may also indicate the size of the graphics buffer. For example, the hardware module **108** may determine that the buffer size associated with the first resolution is unavailable, but that a graphics buffer has been allocated at some other resolution (e.g., 720p using 12 MB of 3 MB continuous blocks). In another example, the buffer response **118** may indicate that the graphics buffer includes a total amount of memory in the appropriate blocks to support a higher resolution than the first resolution (e.g., 84 MB in 21 MB continuous blocks). Then, the playback module **106** may store that information for later use.

[0031] At step **111**, the playback module **106** may cause the media **114** to be displayed by the display **120** at the first resolution (provided that the graphics buffer is of an adequate size). To do so, the playback module **106** may request some or all of the data associated with the media **114** via the network **112**. The network module **104** may receive and transmit the data to the playback module **106**. The playback module **106** may store some or all of the data in the graphics buffer within the RAM **110**. The playback module **106** may then render the data and cause the media **114** to be displayed on the display **120**.

[0032] At step **113**, the playback module **106** may receive a trigger to increase the resolution of the media **114** from the first resolution to a higher, second resolution (e.g., 1080p to 4K). The trigger may include an increase in the available bandwidth of the network **112**. For example, the computing system **102** may have initiated the playback of the media **114** in a first area, where the bandwidth of the network **112** is limited (e.g., a rural area). At some point during the playback of the media **114**, the computing system **102** may travel to a second area with more available bandwidth (e.g., and urban environment). The network module **104** may detect this increase in available bandwidth and notify the playback module **106** of the increase. The playback module **106** may then determine that the new available bandwidth is sufficient to support the second resolution. In other embodiments, the trigger may include a user input, an alert from other components of the computing system **102**, or any other suitable trigger.

[0033] At **115**, the playback module **106** may transmit a second buffer query to the hardware module. The second buffer query **116***b* may be similar to the first buffer query **116***a*, and indicate one or more buffer sizes including a buffer size associated with the second resolution (e.g., 84 MB in 21 MB continuous blocks). In other embodiments, the first buffer response **118***a* may have indicated that the buffer size associated with the second resolution was previously allocated. Then, the playback module **106** may cause the media **114** to be displayed at the second resolution by the display **120**, as described above in relation to the first resolution.

[0034] At **117**, the playback module **106** may receive a second buffer response **118***b*. The second buffer response **118***b* may indicate that the RAM **110** has the buffer size associated with the second resolution available and/or that the graphics buffer has been updated such that the graphics buffer has been allocated. The second buffer response **118***b* may also indicate addresses of blocks within the updated graphics buffer.

[0035] If the second buffer response **118***b* indicates that the graphics buffer has been updated, at step **119**, the playback module **106** may cause the media **114** to be displayed at the second resolution by the display **120**. The playback module **106** may request data associated with the media **114** at the second resolution from the network **112** via the network module **104**. Upon receiving the data, the playback module **106** may cause some or all of the data in the graphics buffer within the RAM **110**, and cause the display **120** to playback the media **114** at the second resolution. If the second buffer response **118***b* indicates that the graphics buffer may not be increased, the playback module **106** may continue to cause the media **114** to be displayed according to the first resolution.

[0036] FIG. **2** illustrates a system **200** including a computing system **202** and networks **212***a-b*, according to certain embodiments. The computing system **202** may be similar to the computing system **102** in FIG. **1** and include similar components and functionalities. The networks **212***a-b* may be a wireless network such as a cellular network (e.g., 3G, 4G, 5G, etc.), WiFi, or other such network. The networks **212***a-b* may be a single network administered by a single network provider. In that case, the first network **212***a* may represent a first connection in a first area (e.g., a rural area). The second network **212***b* may represent a second connection in a second area (e.g., an urban area). In other embodiments, the networks **212***a-b* may be different networks. For example, the first network **212***a* may be associated with a first network provider. The computing system **202** may be roaming utilizing the first network **212***a*, or may be connected to the first network **212***a* by some other arrangement with a second network provider. The second network **212***b* may then be associated with the second network provider. Other possibilities and configurations are readily apparent.

[0037] The computing system **202** may receive a request to stream media while connected to the first network **212***a*. The computing system **202** may determine that a bandwidth **203***a* is available via the network **212***a*. The bandwidth **203***a* may be sufficient to provide the requested media at a first resolution, but not at a higher resolution. For example, the bandwidth **203***a* may be sufficient to support display of the requested media at 720p, but at no higher resolutions. Thus, using some or all of a process similar to the process **101** in FIG. **1**, the computing system **202** may then begin to display the requested media at the first resolution (e.g., 720p).

[0038] During playback of the requested media, the computing system **202** may then connect to the second network **212***b*. The computing system **202** may determine that a bandwidth **203***b* is available via the second network **212***b*. For example, the computing system **202** may be on a train or some other mode of transportation. The computing system **202** may therefore travel within range of the second network **212***b*. In another example, the networks **212***a-b* may be the same network in the same location. Then, the second network **212***b* may represent an increase in available bandwidth from the bandwidth **203***a* to the bandwidth **203***b* due to changes in the networks **212***a-b*, the computing system **202** (e.g., another process began using less bandwidth than previously), or any other reason. The computing system **202** may then determine that the bandwidth **203***b* is sufficient to support a second resolution, higher than the first resolution (e.g., 1080p, 4K, 8K, etc.).

[0039] FIG. **3** illustrates a system **300** for reducing memory fragmentation during media playback, according to certain embodiments. The system **300** may be similar to some or all of the system **100** in FIG. **1** and/or the system **200** in FIG. **2**. The system **300** may include a computing system **302**. The computing system **302** may include a playback module **304**, a hardware module **308**, a processor (s) **310**, and RAM **312**. The components of the computing system **302** should be understood to have similar properties and functions as similar components in the systems **100** and **200**. The computing system **302** may be a unitary device, such as a mobile phone tablet, laptop, etc. Alternatively, some of the components of the computing system **302** may be separate from other components. For example, a display such as monitor, display, or other computing device, connected to the computing system **302** via a wired or wireless connection (e.g., WiFi, Bluetooth, etc.).

[0040] The computing system **302** may receive a request to stream media **318** while connected to a first network (e.g., the network **212***a*). The request may be a user request to stream the media **114** via the first network. The request may be received via the playback module **304** and/or may be transmitted to the playback module **304** via another component of the computing system **302**, such as an operating system. The request may include a desired resolution. The desired resolution may the maximum resolution of the display, or may be a preset resolution based on a predetermined setting. In some embodiments, the playback module **304** may determine a bandwidth needed to support the desired resolution via a lookup table **305**. The lookup table **305** may be included in the playback module **304** (e.g., in code of a software component of the playback module **304**), or may be included in memory of the computing system **302**. The playback module **304** may then query a component of the computing system **302** to determine if the required bandwidth is available (e.g., the network module **104** in FIG. **1**). In some embodiments, the playback module **304** may determine the available bandwidth, then determine a highest possible resolution based on the available bandwidth.

[0041] The playback module **304** may generate a buffer query **316** based at least in part on the available bandwidth and/or data included in the lookup table **305**. For example, the available bandwidth may be sufficient to support display of the media **318** at 1080p. The playback module **304** may then determine that 1080p is associated with a graphics buffer of 24 MB in continuous blocks of 6 MB using the lookup table **305**. The playback module **304** may then generate the buffer query **316** to indicate a graphics buffer with a buffer size of 24 MB in continuous blocks of 6 MB. In other embodiments, the buffer query **316** may include multiple buffer sizes, even if the available bandwidth may not be sufficient to support higher resolutions. For example, the playback module **304** may indicate a first buffer size corresponding to a 4K resolution, a second buffer size corresponding to a 1080p resolution, and a third buffer size corresponding to a 720p resolution. Other possibilities and configuration would be obvious to one of ordinary skill in the art.

[0042] The playback module **304** may then transmit the buffer query **316** to the hardware module **308**. The buffer query **316** may be transmitted using an API or any other suitable method. The hardware module **308** may access one or more components of the computing system **302** to determine a current status of the one or more components. For example, the hardware module **308** may query the processor (s) **310** to determine the availability of the processor(s) **310** to perform tasks such as rendering the media **318**, allocate memory within the RAM **312**, and other such tasks.

[0043] The hardware module **308** may additionally or alternatively determine a status of the RAM **312**. For example, the hardware module **308** may determine that the RAM **312** include blocks **322***a-f*. Each of the blocks **322***a-f* may include the same amount of memory, or may include differing amounts of memory. For example, each of the blocks **322***a-f* may include 2 MB, 6 MB, 30 MB, etc. Each of the blocks **322***a-f* may be further divisible into other continuous blocks. For example, the block **322***b* may include 30 MB. Then, the block **322***b* may be divided to create 4 continuous blocks of up to 7 MB each. The hardware module **308** may then determine an amount of memory indicated for the graphics buffer per the buffer query **316**. To do so, the hardware module **308** may determine that the blocks **322***a*, **322***c*, and **322***e-f* are unavailable. The hardware module **308** may then determine that the blocks **332***b* and **322***d* are available with an amount of memory. For example, each of the blocks **332***b* and **322***d* may include 30 MB of available memory. Thus, the hardware module **308** may determine that the RAM **312** may support a graphics buffer of 24 MB in continuous blocks of 6 MB, as indicated in the buffer query **316**. The hardware module **308** may then cause at the graphics buffer to be allocated within the RAM **312**. The graphics buffer may be allocated in either or both of the blocks **322***b* and **322***d*.

[0044] In embodiments where the buffer query **316** indicates multiple buffer sizes, the hardware module **308** may iteratively determine a maximum buffer size available in the RAM **312**. For example, the buffer query **316** may indicate the first buffer size corresponding to a 4K resolution (e.g., 84 MB in continuous blocks of 21 MB), the second buffer size corresponding to a 1080p resolution (e.g., 24 MB in continuous blocks of 6 MB), and a third buffer size corresponding to a 720p resolution (e.g., 12 MB in continuous blocks of 3 MB). The hardware module **408** may first determine whether or not the RAM **312** includes blocks of continuous memory to support a graphics buffer according to the first buffer size. Here, the blocks **322***b* and **322***d* may each include 30 MB. Thus, the RAM **312** may not support the graphics buffer according to the first buffer size, as only two blocks of 21 MB are available. The hardware module **408** may then determine if the RAM **312** includes blocks of continuous memory to support a graphics buffer according to the second buffer size. Here, the blocks **322***b* and **322***d* may be allocated to create 4 continuous blocks of memory of 6 MB each. The hardware module **308** may therefore not determine whether the RAM **312** can support the third buffer size, as the second buffer size is already supported.

[0045] The hardware module **308** may then transmit a buffer response **320** to the playback module **304** (e.g., via the API). The buffer response **320** may indicate that the graphics buffer has been allocated in the RAM **312** according to a buffer size indicated in the buffer query **316**. The buffer response may also include addresses associated with the blocks **322***b* and **322***d* (and/or subdivisions thereof). In some embodiments, the buffer response **318** may indicate that a graphics buffer has been allocated that is larger than the requested graphics buffer (e.g., that a graphics buffer sufficient to support 4K has been allocated where the request resolution is 1080p.) The playback module **304** may create an entry indicating the larger graphics buffer in the lookup table **305**.

[0046] The playback module **304** may then request the media **318** at the first resolution (e.g., via a network module connected to a network). The playback module **304** may then cause the at least a portion of the media **318** (or data associated with the media **318**), to become buffered data **322**. The playback module **304** may cause the buffered data **322** to be stored within the graphics buffer of the RAM **312**. The playback module **304** may then cause the media **318** to be displayed at the first resolution (here, 1080p). The playback module **304** may retrieve and render the buffered data **322** and/or render other data associated with the media **318**. The media **318** may then be displayed on the display.

[0047] FIG. **4** illustrates system **400** for reducing memory fragmentation during media playback, according to certain embodiments. The system **400** may be similar to the system **300** and include similar components and functionalities. In some embodiments, the system **400** may be the same system **300**, connected to a different network (e.g., the computing system **200** connecting to the networks **312***a-b*). The system **400** may also some or all of the system **100** in FIG. **1** and/or the system **200** in FIG. **2**. The system **400** may include a computing system **402**. The computing system **402** may include a playback module **404**, a hardware module **408**, a processor(s) **410**, and RAM **412**. The components of the computing system **402** should be understood to have similar properties and functions as similar components in the systems **100** and **200**. The computing system **402** may be a unitary device, such as a mobile phone tablet, laptop, etc. Alternatively, some of the components of the computing system **402** may be separate from other components. For example, a display such as monitor, display, or other computing device, connected to the computing system **402** via a wired or wireless connection (e.g., WiFi, Bluetooth, etc.).

[0048] The computing system **402** and/or the playback module **404** may receive a trigger **414**. The trigger **414** may indicate an increase in available bandwidth over a network, such as the bandwidth **213***b* in FIG. **2**. The trigger **414** may indicate a user input, a change in the status of a component of the computing system **402**, or any other appropriate trigger. The trigger **414** may also indicate a higher resolution. Continuing the example shown in FIG. **3**, the trigger **414** may indicate a request to display media **418** at a 4K resolution, where the computing system **402** is currently displaying the media **418** at 1080p.

[0049] In response to the trigger **414**, the playback module **404** may determine a highest possible resolution according to the available bandwidth (e.g., by using a lookup table **405**). The playback module **404** may also determine a buffer size associated with the highest possible resolution. The playback module **404** may then generate a buffer query **416**, indicating the buffer size associated with the higher resolution. For example, the available bandwidth may be sufficient to support display of the media **418** at 4K. The playback module **404** may then determine that 4K is associated with a graphics buffer of 84 MB in continuous blocks of 21 MB using the lookup table **405**. The playback module **404** may then generate the buffer query **416** to indicate a graphics buffer with a buffer size of 84 MB in continuous blocks of 21 MB.

[0050] The playback module **404** may then transmit the buffer query **416** to the hardware module **408**. The buffer query **416** may be transmitted using an API or any other suitable method. The hardware module **408** may access one or more components of the computing system **402** to determine a current status of the one or more components. For example, the hardware module **408** may query the processor (s) **410** to determine the availability of the processor(s) **410** to perform tasks such as rendering the media **418**, allocate memory within the RAM **412**, and other such tasks.

[0051] The hardware module **408** may additionally or alternatively determine a status of the RAM **412**. For example, the hardware module **408** may determine that the RAM **412** includes blocks **422***a-f*. Each of the blocks **422***a-f* may include the same amount of memory or may include differing amounts of memory. For example, each of the blocks **422***a-f* may include 2 MB, 6 MB, 40 MB, etc. Each of the blocks **422***a-f* may be further divisible into other continuous blocks. For example, the block **422***a* may include 30 MB. Then, the block **422***a* may be divided to create up to 4 continuous blocks of up to 7 MB each. The block **422***a* may also be used to allocate a single continuous block of 21 MB. The hardware module **408** may then determine an amount of memory indicated for the graphics buffer per the buffer query **416**. To do so, the hardware module **408** may determine that the blocks **422***b*, and **422***e* are unavailable. The hardware module **408** may then determine that the blocks **442***a*, **422***c-d*, and **422***f* are available with an amount of memory. For example, each of the blocks **442***a*, **422***c-d*, and **422***f* may include 30 MB of available memory. Thus, the hardware module **408** may determine that the RAM **412** may support a graphics buffer of 84 MB in continuous blocks of 21 MB, as indicated in the buffer query **416**. The hardware module **408** may then cause at the graphics buffer to be allocated within the RAM **412**. The graphics buffer may be allocated in using each of the blocks **442***a*, **422***c-d*, and **422***f*.

[0052] The hardware module **408** may then transmit a buffer response **420** to the playback module **404** (e.g., via the API). The buffer response **420** may indicate that the graphics buffer has been allocated in the RAM **412** according to a buffer size indicated in the buffer query **416**. The buffer response may also include addresses associated with the blocks **442***a*, **422***c-d*, and **422***f* (and/or subdivisions thereof). In some embodiments, the hardware module **408** may determine that the RAM **412** may not support the higher resolution. Then, the buffer response **420** may indicate that the higher resolution is not supportable, and the playback module **404** may continue to display the media **418** at the first resolution (e.g., 1080p as described in FIG. **3**).

[0053] If the buffer response **420** indicates that the graphics buffer may support the higher resolution, the playback module **404** may then request the media **418** at the higher resolution (e.g., via a network module connected to a network). The playback module **404** may then cause the at least a portion of the media **418** (or data associated with the media **418**), to become buffered data **422**. The playback module **404** may cause the buffered data **422** to be stored within the graphics buffer of the RAM **412**. The playback module **404** may then cause the media **418** to be displayed at the higher resolution (here, 4K). The playback module **404** may retrieve and render the buffered data **422** and/or render other data associated with the media **418**. The media **418** may then be displayed on the display.

[0054] FIG. **5** illustrates a flowchart of a method **500** for reducing memory fragmentation during media playback, according to certain embodiments. The method **500** may be performed by some or all of the systems and devices described herein. For example, the method **500** may be performed by the systems **100**, **200**, **300**, and **400**, alone or in working in conjunction, in FIGS. **1**-**4**, respectively. In some embodiments, the steps of the method **500** may be performed in a different order than is described, and/or may be combined with other steps. In some embodiments, some steps of the method **500** may be skipped altogether.

[0055] At step **502**, the method **500** may include receiving, by a playback module of a computing system, a trigger to increase a resolution of media to be displayed from a first resolution to a second resolution. The computing system may be similar to the computing system **102** in FIG. **1** and/or the computing system **400** in FIG. **4**. The playback module may therefore be similar to the playback module **106** and/or **404**. The trigger may indicate an increase in available bandwidth over a network, such as the bandwidth **213***b* in FIG. **2**. The trigger may indicate a user input, a change in the status of a component of the computing system, or any other appropriate trigger. The trigger may also indicate the second resolution. For example, the first resolution may be 720p, and the second resolution may be 4K. The computing system may be displaying the media at the first resolution, or the trigger may initiate the computing system to begin displaying the media. The playback module may determine the first resolution based on an available bandwidth, a status of one or more components of the computing system, and/or other factors.

[0056] At step **504**, the method **500** may include determining, by the playback module of the computing system, a minimum buffer required to display the media at the second resolution. The playback module may determine the minimum buffer based at least in part on a lookup table such as the lookup table **405**. The minimum buffer may be a total amount of memory needed in continuous blocks in order to support playback of the media at the second resolution.

[0057] At step **506**, the method **500** may include transmitting, by the playback module of the computing system, a buffer query to a hardware module of the computing system. The buffer query indicating the minimum buffer. The buffer query may be similar to the buffer query **316** and/or the buffer query **416**. The buffer query may be based at least in part on the available bandwidth and/or data included in the lookup table. For example, the available bandwidth may be sufficient to support display of the media at 1080p. The playback module may then determine that 1080p is associated with a graphics buffer of 24 MB in continuous blocks of 6 MB using the lookup table. The playback module may then generate the buffer query to indicate a graphics buffer with a buffer size of 24 MB in continuous blocks of 6 MB. In other embodiments, the buffer query may include multiple buffer sizes, even if the available bandwidth may not be sufficient to support higher resolutions. For example, the playback module may indicate a first buffer size corresponding to a 4K resolution, a second buffer size corresponding to a 1080p resolution, and a third buffer size corresponding to a 720p resolution. Other possibilities and configuration would be obvious to one of ordinary skill in the art.

[0058] The playback module may then transmit the buffer query to the hardware module. The buffer query may be transmitted using an API or any other suitable method. The hardware module may access one or more components of the computing system to determine a current status of the one or more components. For example, the hardware module may query a processor(s) to determine the availability of the processor(s) to perform tasks such as rendering the media, allocate memory within a RAM, and other such tasks.

[0059] At step **508**, the method **500** may include receiving, by the playback module of the computing system, a buffer response from the hardware module of the computing system. The buffer response may indicate whether the minimum buffer is available. The buffer response may be similar to the buffer response **320** and/or the buffer response **320**. The buffer response may also include an address associated with the minimum buffer and/or continuous blocks of memory thereof. The buffer response may also indicate that a graphics buffer has been allocated within the RAM according to the buffer query.

[0060] In response to the buffer response indicating that the minimum buffer is available At step **510**, the method **500** may include causing, by the playback module of the computing system, the media to be displayed at the second resolution. The media may be displayed on a display included in the computing system, or may be displayed on a connected display. The playback module may request the media at the second resolution (e.g., via a network module connected to a network). The playback module may then cause the at least a portion of the media (or data associated with the media), to become buffered data. The playback module may cause the buffered data to be stored within the graphics buffer of the RAM. The playback module may then cause the media to be displayed at the second resolution (e.g., 4K). The playback module may retrieve and render the buffered data and/or render other data associated with the media. The media may then be displayed on the display.

[0061] In some embodiments, the buffer response may indicate that the minimum buffer is not available. Then, in response to the buffer response indicating that the minimum buffer is unavailable, the method **500** may include causing, by the playback module of the computing system, the media to be displayed at the first resolution.

[0062] In some embodiments, the method **500** may include determining, by the computing system, a first bandwidth for wireless communications via a wireless network. The first bandwidth may be similar to the bandwidth **213***a* in FIG. **2**. The wireless network may be a cellular network, WiFi, or any other suitable network. The method **500** may then include detecting, by the computing system, an increase in the first bandwidth for wireless communications to a second bandwidth (e.g., the bandwidth **213***b* in FIG. **2**). The method **500** may also include determining, by the playback module of the computing system, that the second bandwidth is above a certain threshold such that the media may be displayed at the second resolution.

[0063] FIG. **6** illustrates an exemplary computer system **600**, according to certain embodiments. The computer system **600** may be used to implement any of the methods described herein. For example, the system **600** may at least partially control a device for the playback of media and related systems to perform the method **500**. As shown in the figure, computer system **600** includes a processing unit **604** that communicates with a number of peripheral subsystems via a bus subsystem **602**. These peripheral subsystems may include a processing acceleration unit **606**, an I/O subsystem **608**, a storage subsystem **618** and a communications subsystem **624**. Storage subsystem **618** includes tangible computer-readable storage media **622** and a system memory **610**.

[0064] Bus subsystem **602** provides a mechanism for letting the various components and subsystems of computer system **600** communicate with each other as intended. Although bus subsystem **602** is shown schematically as a single bus, alternative embodiments of the bus subsystem may utilize multiple buses. Bus subsystem **602** may be any of several types of bus structures including a memory bus or memory controller, a peripheral bus, and a local bus using any of a variety of bus architectures. For example, such architectures may include an Industry Standard Architecture (ISA) bus, Micro Channel Architecture (MCA) bus, Enhanced ISA (EISA) bus, Video Electronics Standards Association (VESA) local bus, and Peripheral Component Interconnect (PCI) bus, which can be implemented as a Mezzanine bus manufactured to the IEEE P1386.1 standard.

[0065] Processing unit **604**, which can be implemented as one or more integrated circuits (e.g., a conventional microprocessor or microcontroller), controls the operation of computer system **600**. One or more processors may be included in processing unit **604**. These processors may include single core or multicore processors. In certain embodiments, processing unit **604** may be implemented as one or more independent processing units **632** and/or **634** with single or multicore processors included in each processing unit. In other embodiments, processing unit **604** may also be implemented as a quad-core processing unit formed by integrating two dual-core processors into a single chip.

[0066] In various embodiments, processing unit **604** can execute a variety of programs in response to program code and can maintain multiple concurrently executing programs or processes. At any given time, some or all of the program code to be executed can be resident in processor(s) **604** and/or in storage subsystem **618**. Through suitable programming, processor(s) **604** can provide various functionalities described above. Computer system **600** may additionally include a processing acceleration unit **606**, which can include a digital signal processor (DSP), a special-purpose processor, and/or the like.

[0067] I/O subsystem **608** may include user interface input devices and user interface output devices. User interface input devices may include a keyboard, pointing devices such as a mouse or trackball, a touchpad or touch screen incorporated into a display, a scroll wheel, a click wheel, a dial, a button, a switch, a keypad, audio input devices with voice command recognition systems, microphones, and other types of input devices. User interface input devices may include, for example, motion sensing and/or gesture recognition devices that enables users to control and interact with an input device through a natural user interface using gestures and spoken commands. Additionally, user interface input devices may include voice recognition sensing devices that enable users to interact with voice recognition systems through voice commands.

[0068] User interface input devices may also include, without limitation, three dimensional (3D) mice, joysticks or pointing sticks, gamepads and graphic tablets, and audio/visual devices such as speakers, digital cameras, digital camcorders, portable media players, webcams, image scanners, fingerprint scanners, barcode reader, 3D scanners, 3D printers, laser rangefinders, and eye gaze tracking devices. Additionally, user interface input devices may include, for example, medical imaging input devices such as computed tomography, magnetic resonance imaging, position emission tomography, medical ultrasonography devices. User interface input devices may also include, for example, audio input devices such as MIDI keyboards, digital musical instruments and the like.

[0069] User interface output devices may include a display subsystem, indicator lights, or non-visual displays such as audio output devices, etc. The display subsystem may be a cathode ray tube (CRT), a flat-panel device, such as that using a liquid crystal display (LCD) or plasma display, a projection device, a touch screen, and the like. In general, use of the term "output device" is intended to include all possible types of devices and mechanisms for outputting information from computer system **600** to a user or other computer. For example, user interface output devices may include, without limitation, a variety of display devices that visually convey text, graphics and audio/video information such as monitors, printers, speakers, headphones, automotive navigation systems, plotters, voice output devices, and modems.

[0070] Computer system **600** may comprise a storage subsystem **618** that comprises software elements, shown as being currently located within a system memory **610**. System memory **610** may store program instructions that are loadable and executable on processing unit **604**, as well as data generated during the execution of these programs.

[0071] Depending on the configuration and type of computer system **600**, system memory **610** may be volatile (such as random access memory (RAM)) and/or non-volatile (such as read-only memory (ROM), flash memory, etc.). The RAM typically contains data and/or program modules that are immediately accessible to and/or presently being operated and executed by processing unit **604**. In some implementations, system memory **610** may include multiple different types of memory, such as static random access memory (SRAM) or dynamic random access memory (DRAM). In some implementations, a basic input/output system (BIOS), containing the basic routines that help to transfer information between elements within computer system **600**, such as during start-up, may typically be stored in the ROM. By way of example, and not limitation, system memory **610** also illustrates application programs **612**, which may include client applications, Web browsers, mid-tier applications, relational database management systems (RDBMS), etc., program data **614**, and an operating system **616**.

[0072] Storage subsystem **618** may also provide a tangible computer-readable storage medium for storing the basic programming and data constructs that provide the functionality of some embodiments. Software (programs, code modules, instructions) that when executed by a processor provide the functionality described above may be stored in storage subsystem **618**. These software modules or instructions may be executed by processing unit **604**. Storage subsystem **618** may also provide a repository for storing data used in accordance with some embodiments.

[0073] Storage subsystem **618** may also include a computer-readable storage media reader **620** that can further be connected to computer-readable storage media **622**. Together and, optionally, in combination with system memory **610**, computer-readable storage media **622** may comprehensively represent remote, local, fixed, and/or removable storage devices plus storage media for temporarily and/or more permanently containing, storing, transmitting, and retrieving computer-readable information.

[0074] Computer-readable storage media **622** containing code, or portions of code, can also include any appropriate media, including storage media and communication media, such as but not limited to, volatile and non-volatile, removable and non-removable media implemented in any method or technology for storage and/or transmission of information. This can include tangible computer-readable storage media such as RAM, ROM, electronically erasable programmable ROM (EEPROM), flash memory or other memory technology, CD-ROM, digital versatile disk (DVD), or other optical storage, magnetic cassettes, magnetic tape, magnetic disk storage or other magnetic storage devices, or other tangible computer readable media. This can also include nontangible computer-readable media, such as data signals, data transmissions, or any other medium which can be used to transmit the desired information and which can be accessed by computing system **600**.

[0075] By way of example, computer-readable storage media **622** may include a hard disk drive that reads from or writes to non-removable, nonvolatile magnetic media, a magnetic disk drive that reads from or writes to a removable, nonvolatile magnetic disk, and an optical disk drive that reads from or writes to a removable, nonvolatile optical disk such as a CD ROM, DVD or other optical media. Computer-readable storage media **622** may include, but is not limited to, flash memory cards, universal serial bus (USB) flash drives, secure digital (SD) cards, DVD disks, digital video tape, and the like. Computer-readable storage media **622** may also include, solid-state drives (SSD) based on non-volatile memory such as flash-memory based SSDs, enterprise flash drives, solid state ROM, and the like, SSDs based on volatile memory such as solid state RAM, dynamic RAM, static RAM, DRAM-based SSDs, magnetoresistive RAM (MRAM) SSDs, and hybrid SSDs that use a combination of DRAM and flash memory based SSDs. The disk drives and their associated computer-readable media may provide non-volatile storage of computer-readable instructions, data structures, program modules, and other data for computer system **600**.

[0076] Communications subsystem **624** provides an interface to other computer systems and networks. Communications subsystem **624** serves as an interface for receiving data from and transmitting data to other systems from computer system **600**. For example, communications subsystem **624** may enable computer system **600** to connect to one or more devices via the Internet. In some embodiments communications subsystem **624** can include radio frequency (RF) transceiver components for accessing wireless voice and/or data networks (e.g., using cellular telephone technology, advanced data network technology, such as 3G, 4G, 5G, or EDGE (enhanced data rates for global evolution), WiFi (IEEE 802.4 family standards, or other mobile communication technologies, or any combination thereof), global positioning system (GPS) receiver components, and/or other components. In some embodiments communications subsystem **624** can provide wired network connectivity (e.g., Ethernet) in addition to or instead of a wireless interface.

[0077] In some embodiments, communications subsystem **624** may also receive input communication in the form of structured and/or unstructured data feeds **626** such as the media **118** in FIG. **1**, event streams **628**, event updates **630**, and the like on behalf of one or more users who may use computer system **600**.

[0078] By way of example, communications subsystem **624** may be configured to receive data feeds **626** in real-time from users of social networks and/or other communication services, web feeds such as Rich Site Summary (RSS) feeds, and/or real-time updates from one or more third party information sources.

[0079] Additionally, communications subsystem **624** may also be configured to receive data in the form of continuous data streams, which may include event streams **628** of real-time events and/or event updates **630**, that may be continuous or unbounded in nature with no explicit end. Examples of applications that generate continuous data may include, for example, sensor data applications, financial tickers, network performance measuring tools (e.g., network monitoring and traffic management applications), clickstream analysis tools, automobile traffic monitoring, and the like.

[0080] Communications subsystem **624** may also be configured to output the structured and/or unstructured data feeds **626**, event streams **628**, event updates **630**, and the like to one or more databases that may be in communication with one or more streaming data source computers coupled to computer system **600**.

[0081] Due to the ever-changing nature of computers and networks, the description of computer system **600** depicted in the figure is intended only as a specific example. Many other configurations having more or fewer components than the system depicted in the figure are possible. For example, customized hardware might also be used and/or particular elements might be implemented in hardware, firmware, software (including applets), or a combination. Further, connection to other computing devices, such as network input/output devices, may be employed. Based on the disclosure and teachings provided herein, other ways and/or methods to implement the various embodiments should be apparent.

[0082] The methods, systems, and devices discussed above are examples. Some embodiments were described as processes depicted as flow diagrams or block diagrams. Although each may describe the operations as a sequential process, many of the operations can be performed in parallel or concurrently. In addition, the order of the operations may be rearranged. A process may have additional steps not included in the figure. It will be further appreciated that all testing methods described here may be based on the testing standards in use at the time of filing or those developed after filing.

[0083] It should be noted that the systems and devices discussed above are intended merely to be examples. It must be stressed that various embodiments may omit, substitute, or add various procedures or components as appropriate. Also, features described with respect to certain embodiments may be combined in various other embodiments. Different aspects and elements of the embodiments may be combined in a similar manner. Also, it should be emphasized that technology evolves and, thus, many of the elements are examples and should not be interpreted to limit the scope of the invention.

[0084] Specific details are given in the description to provide a thorough understanding of the embodiments. However, it will be understood by one of ordinary skill in the art that the embodiments may be practiced without these specific details. For example, well-known structures and techniques have been shown without unnecessary detail in order to avoid obscuring the embodiments. This description provides example embodiments only, and is not intended to limit the scope, applicability, or configuration of the invention. Rather, the preceding description of the embodiments will provide those skilled in the art with an enabling description for implementing embodiments of the invention. Various changes may be made in the function and arrangement of elements without departing from the spirit and scope of the invention.

[0085] Having described several embodiments, it will be recognized by those of skill in the art that various modifications, alternative constructions, and equivalents may be used without departing from the spirit of the invention. For example, the above elements may merely be a component of a larger system, wherein other rules may take precedence over or otherwise modify the application of the invention. Also, a number of steps may be undertaken before, during, or after the above elements are considered. Accordingly, the above description should not be taken as limiting the scope of the invention.

[0086] Also, the words "comprise", "comprising", "contains", "containing", "include", "including", and "includes", when used in this specification and in the following claims, are intended to specify the presence of stated features, integers, components, or steps, but they do not preclude the presence or addition of one or more other features, integers, components, steps, acts, or groups.

[0087] Unless defined otherwise, all technical and scientific terms used herein have the same meaning as commonly or conventionally understood. As used herein, the articles "a" and "an" refer to one or to more than one (i.e., to at least one) of the grammatical object of the article. By way of example, "an element" means one element or more than one element. "About" and/or "approximately" as used herein when referring to a measurable value such as an amount, a temporal duration, and the like, encompasses variations of ±20% or ±10%, ±5%, or ±0.1% from the specified value, as such variations are appropriate to in the context of the systems, devices, circuits, methods, and other implementations described herein. "Substantially" as used herein when referring to a measurable value such as an amount, a temporal duration, a physical attribute (such as frequency), and the like, also encompasses variations of ±20% or ±10%, ±5%, or ±0.1% from the specified value, as such variations are appropriate to in the context of the systems, devices, circuits, methods, and other implementations described herein.

[0088] As used herein, including in the claims, "and" as used in a list of items prefaced by "at least one of" or "one or more of" indicates that any combination of the listed items may be used. For example, a list of "at least one of A, B, and C" includes any of the combinations A or B or C or AB or AC or BC and/or ABC (i.e., A and B and C). Furthermore, to the extent more than one occurrence or use of the items A, B, or C is possible, multiple uses of A, B, and/or C may form part of the contemplated combinations. For example, a list of "at least one of A, B, and C" may also include AA, AAB, AAA, BB, etc.

What is claimed is:

1. A method, comprising:

receiving, by a playback module of a computing system, a trigger to increase a resolution of media to be displayed from a first resolution to a second resolution;

determining, by the playback module of the computing system, a minimum buffer associated with the second resolution;

transmitting, by the playback module of the computing system, a buffer query to a hardware module of the computing system, the buffer query indicating the minimum buffer;

receiving, by the playback module of the computing system, a buffer response from the hardware module of the computing system, the buffer response indicating whether the minimum buffer is available; and

in response to the buffer response indicating that the minimum buffer is available:

causing, by the playback module of the computing system, the media to be displayed at the second resolution.

2. The method of claim 1, further comprising:

in response to the buffer response indicating that the minimum buffer is unavailable:

causing, by the playback module of the computing system, the media to be displayed at the first resolution.

3. The method of claim 2, wherein the first resolution is at least one of 1080p or 720p, and the second resolution is 4K.

4. The method of claim 1, wherein the minimum buffer corresponds to at least one continuous block of random access memory.

5. The method of claim 1, wherein the buffer query is transmitted to the hardware module of the computing system using an application programming interface.

6. The method of claim 1, further comprising:

determining, by the computing system, a first bandwidth for wireless communications via a wireless network;

detecting, by the computing system, an increase in the first bandwidth for wireless communications to a second bandwidth; and

determining, by the playback module of the computing system, that the second bandwidth is above a certain threshold such that the media may be displayed at the second resolution.

7. The method of claim 6, wherein the computing system transmits the buffer request in response to determining that the second bandwidth is above the certain threshold.

8. The method of claim 1, wherein the trigger comprises a change in bandwidth available to the computing system via a network.

9. The method of claim 1, further comprising:

in response to the buffer response indicating that the minimum buffer is available:

receiving, from the hardware module of the computing system, a location of a continuous block of memory within a computer-readable memory corresponding to the minimum buffer,

causing, by the playback module, data associated with the media to be stored in at least a portion of the continuous block of memory.

10. A system, comprising:

one or more processors;

a hardware module, configured to access data indicating a status of one or more hardware components of the system;

a playback module, configured to cause media to be displayed by the system; and

a non-transitory computer-readable medium comprising instructions that, when executed by the system, cause the system to perform operations to:

receive, by the playback module, a trigger to increase a resolution of media to be displayed from a first resolution to a second resolution;

determine, by the playback module, a minimum buffer associated with the second resolution;

transmit, by the playback module, a buffer query to the hardware module, the buffer query indicating the minimum buffer;

receive, by the playback module, a buffer response from the hardware module of the computing system, the buffer response indicating whether the minimum buffer is available; and

in response to the buffer response indicating that the minimum buffer is available:

cause, by the playback module, the media to be displayed at the second resolution.

11. The system of claim 10, wherein the first resolution is at least one of 1080p or 720p, and the second resolution is 4K.

12. The system of claim 10, wherein the buffer query comprises respective buffer sizes associated with at least one of the first resolution or the second resolution.

13. The system of claim 10, wherein the first resolution is determined at least in part by an available bandwidth of a network.

14. The system of claim 10, wherein the system further performs operations to:

determine a first bandwidth for wireless communications via a wireless network;

detect an increase in the first bandwidth for wireless communications to a second bandwidth; and

determine, by the playback module, that the second bandwidth is above a certain threshold such that the media may be displayed at the second resolution.

15. The system of claim 14, wherein the playback module transmits the buffer query in response to determining that the second bandwidth is above the certain threshold.

16. A non-transitory computer-readable medium comprising instructions that, when executed by one or more processors, cause the one or more processors to perform operations comprising:

receiving, by a playback module of a computing system, a trigger to increase a resolution of media to be displayed from a first resolution to a second resolution;

determining, by the playback module of the computing system, a minimum buffer associated with the second resolution;

transmitting, by the playback module of the computing system, a buffer query to a hardware module of the computing system, the buffer query indicating the minimum buffer;

receiving, by the playback module of the computing system, a buffer response from the hardware module of the computing system, the buffer response indicating whether the minimum buffer is available; and

in response to the buffer response indicating that the minimum buffer is available:

causing, by the playback module of the computing system, the media to be displayed at the second resolution.

17. The non-transitory computer-readable medium of claim 16, wherein the minimum buffer corresponds to a plurality of continuous blocks of random access memory.

18. The non-transitory computer-readable medium of claim 16, wherein the buffer query is transmitted to the hardware module of the computing system using an application programming interface.

19. The non-transitory computer-readable medium of claim 16, further comprising:

determining, by the computing system, a first bandwidth for wireless communications via a wireless network;

detecting, by the computing system, an increase in the first bandwidth for wireless communications to a second bandwidth; and

determining, by the playback module of the computing system, that the second bandwidth is above a certain threshold such that the media may be displayed at the second resolution.

20. The non-transitory computer-readable medium of claim **16**, further comprising:

in response to the buffer response indicating that the minimum buffer is available:

receiving, from the hardware module of the computing system, a location of a continuous block of memory within a computer-readable memory corresponding to the minimum buffer,

causing, by the playback module, data associated with the media to be stored in at least a portion of the continuous block of memory.

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